Unit 5 Guided Notes — The Pythagorean Theorem

Name _____ Date _____ Period _____

$a^2 + b^2 = c^2$. Square the legs and add to find c^2. To find a leg, subtract the squares.

🎯 By the end of this unit, I can…

- Use the Pythagorean Theorem to find a missing side of a right triangle
- Apply its converse to test whether a triangle is right
- Solve real-world and coordinate-distance problems

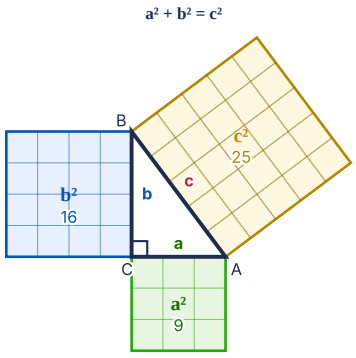
Standards: NC.8.G.6 · NC.8.G.7 · NC.8.G.8

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 The Pythagorean Theorem

🔑 THE THEOREM: In a RIGHT triangle, the sum of the squares of the two LEGS equals the square of the HYPOTENUSE: $a^2 + b^2 = c^2$.

a, b = legs (shorter sides) · c = hypotenuse (longest side, across from the right angle)



The two smaller squares (a^2+b^2) EXACTLY fill the biggest square (c^2)

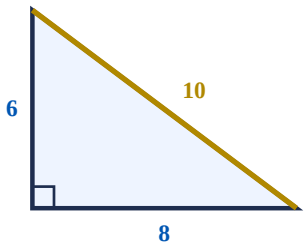
⚠️ **IMPORTANT:** The Pythagorean Theorem ONLY works on RIGHT triangles (triangles with a 90° angle)!

The two shorter sides of a right triangle are called the _____ (labeled a and b). The longest side, directly across from the right angle, is the _____ (labeled c). The theorem written as an equation: $a^2 + b^2 =$ _____. It applies only to _____ triangles.

✏ Practice: The Pythagorean Theorem

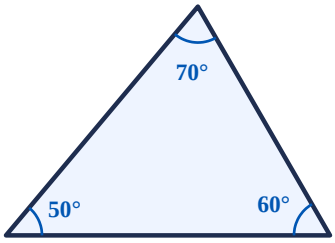
1. ★ Try it Look at the right triangle. Which side is the hypotenuse?

- ○ 6
- ○ 8
- ○ 10



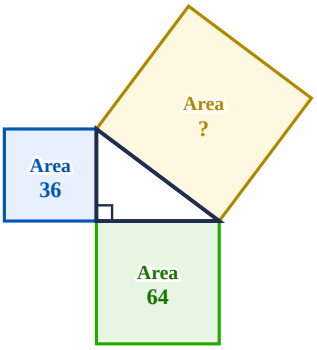
2. ★ Try it A triangle has angles of 50°, 60°, and 70°. Can you use $a^2 + b^2 = c^2$ on it?

- ○ Yes
- ○ No

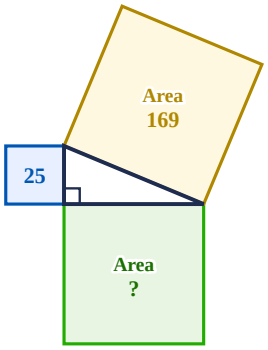


3. ★★ On level The squares on the two legs of a right triangle have areas 36 and 64. What is the area of the square on the hypotenuse?

Answer: _____



4. ★★★ Challenge The square on the hypotenuse has area 169. The square on one leg has area 25. What is the area of the square on the other leg?

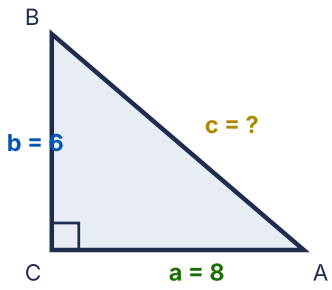


Answer: _____

2 Finding the Hypotenuse

🔑 **WHEN TO USE:** When you know BOTH LEGS and need to find the HYPOTENUSE.
Formula: $c^2 = a^2 + b^2 \rightarrow c = \sqrt{(a^2 + b^2)}$

FIND THE HYPOTENUSE



Example: Legs are 6 and 8. Find the hypotenuse.

Step 1 — Write formula:

Step 2 — Substitute:

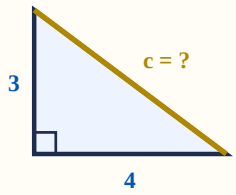
Step 3 — Square:

Step 4 — Add:

✓ **Step 5 — Square root:**

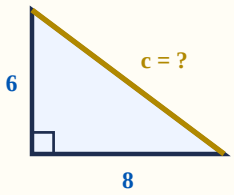
Practice — Find the Hypotenuse (c)

★ Try it Legs 3 and 4: $3^2 + 4^2 = c^2$ → c = ?



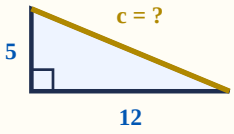
c = _____

★ Try it Legs 6 and 8: $6^2 + 8^2 = c^2$ → c = ?



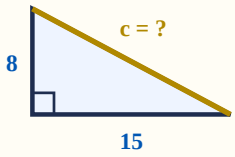
c = _____

★ Try it Legs 5 and 12: $5^2 + 12^2 = c^2$ → c = ?



c = _____

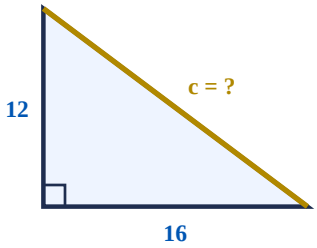
★ Try it Legs 8 and 15: $8^2 + 15^2 = c^2$ → c = ?



c = _____

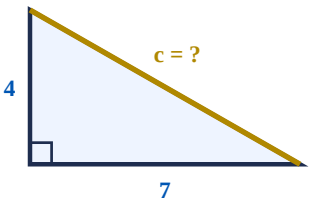
✏ Practice: Finding the Hypotenuse

1. ★ Try it The legs of a right triangle are 12 and 16. Find the hypotenuse c.



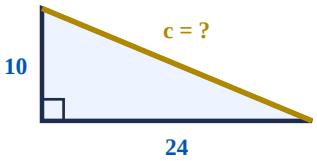
Answer: _____

2. ★★ On level The legs are 4 and 7. Find the hypotenuse c. Round to the nearest tenth (or give the exact √ answer).



Answer: _____

3. ★★★ Challenge The legs are 10 and 24. Maria says c = 34, because 10 + 24 = 34. What is the correct value of c?



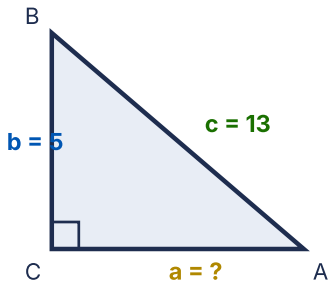
Answer: _____

3 Finding a Missing Leg

🔑 **WHEN TO USE:** When you know ONE LEG and the HYPOTENUSE. Formula: $a^2 = c^2 - b^2 \rightarrow a = \sqrt{(c^2 - b^2)}$

⚠️ **REMEMBER:** Finding a LEG = SUBTRACT from the hypotenuse squared. Finding the HYPOTENUSE = ADD the two legs squared.

FIND THE MISSING LEG



Example: One leg is 5, hypotenuse is 13. Find the other leg.

Step 1 — Write formula:

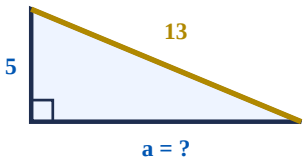
Step 2 — Substitute:

Step 3 — Square:

Step 4 — Subtract 25 (finding a LEG = SUBTRACT):

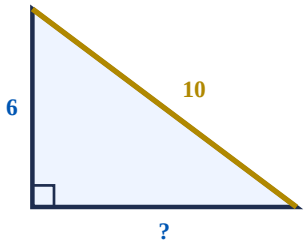
✓ **Step 5 — Square root:**

ERROR ANALYSIS A student finds a missing leg when c = 13, b = 5 by writing $a^2 = 13^2 + 5^2 = 194$. What's wrong?



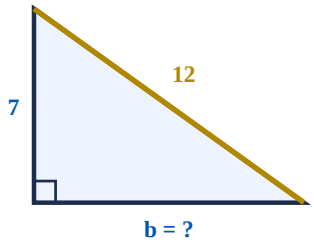
✏ Practice: Finding a Missing Leg

1. ★ Try it One leg is 6 and the hypotenuse is 10. Find the other leg.



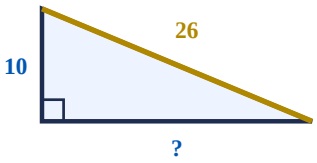
Answer: _____

2. ★★ On level One leg is 7 and the hypotenuse is 12. Find the other leg. Round to the nearest tenth (or give the exact √ answer).



Answer: _____

3. ★★★ Challenge A right triangle has a hypotenuse of 26 and one leg of 10. What is its perimeter?

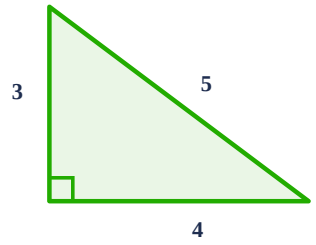


Answer: _____

4 Converse of the Pythagorean Theorem

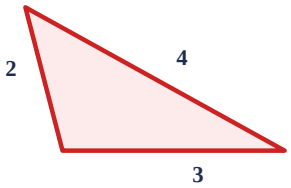
🔑 **THE CONVERSE:** Given three side lengths $a \le b \le c$, test: does $a^2 + b^2 = c^2$? If YES → right triangle. If NO → not a right triangle. Always test the two SMALLER sides against the LARGEST.

3, 4, 5
RIGHT triangle ✓



$3^2 + 4^2 = 25$ and $5^2 = 25$

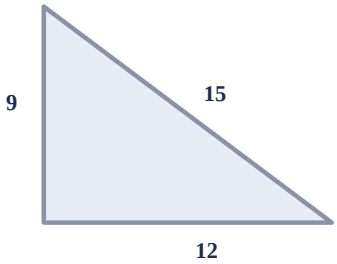
2, 3, 4
NOT a right triangle ×



$2^2 + 3^2 = 13$ but $4^2 = 16$

Example: Sides 9, 12, 15. Is it a right triangle?

9, 12, 15



Step 1 — Order the sides:

Step 2 — Write the test:

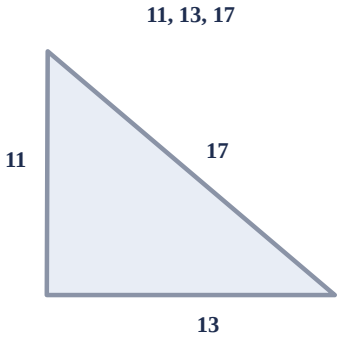
Step 3 — Substitute the two SMALLER sides:

Step 4 — Square each one:

Step 5 — Add the left side:

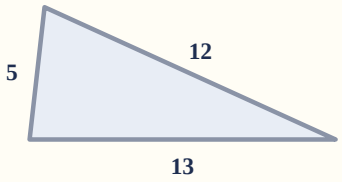
✓ Step 6 — Compare and conclude:

⚠️ **Your eyes cannot do this job.** Every triangle below is drawn to true scale. Look at **11, 13, 17**: it is *not* a right triangle, because $11^2 + 13^2 = 121 + 169 =$ **290** while $17^2 =$ **289**. It misses by one. Drawn accurately it sits two-tenths of a degree off square — nobody alive could spot that. This is exactly why the converse exists: **you must run the arithmetic, every time.**



Quick Checks

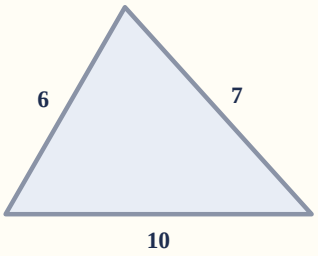
★ Try it **MC** **Sides 5, 12, 13 — right triangle?**



not to scale

- ○ Yes
- ○ No

★ Try it **MC** **Sides 6, 7, 10 — right triangle?**



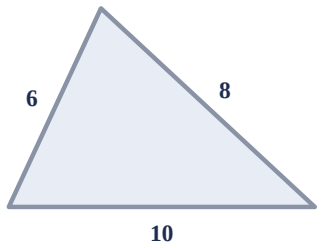
not to scale

- ○ Yes
- ○ No

✏ Practice: Converse of the Pythagorean Theorem

1. ★ Try it A triangle has sides 6, 8, and 10. Is it a right triangle?

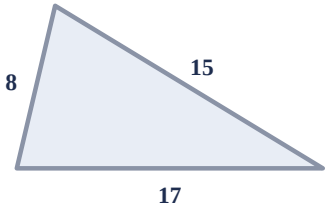
- ○ Yes
- ○ No



not to scale

2. ★★ On level A triangle has sides 17, 8, and 15. Is it a right triangle?

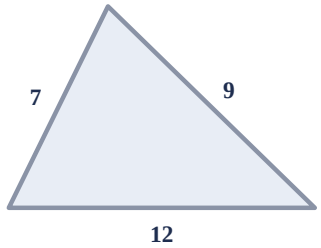
- ○ Yes
- ○ No



not to scale

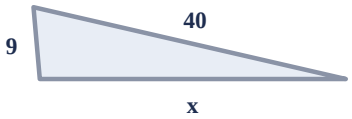
3. ★★ On level A triangle has sides 7, 9, and 12. Is it a right triangle?

- ○ Yes
- ○ No



not to scale

4. ★★★ Challenge A triangle has sides 9, 40, and x. The side x is the longest. What must x be for a right triangle?



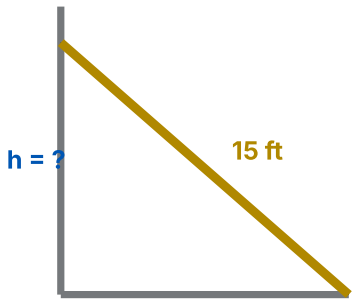
not to scale

Answer: _____

5 Word Problems

📌 **STRATEGY:** (1) Read carefully (2) Draw & label a diagram (3) Identify legs vs hypotenuse (4) Set up $a^2 + b^2 = c^2$ (5) Solve step by step.

LADDER AGAINST A WALL



$9^2 + h^2 = 15^2$

Ladder Against a Wall: A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high does it reach?

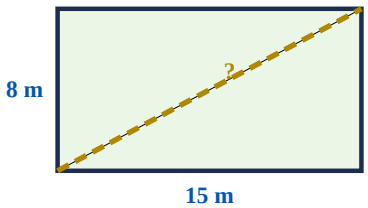
Setup:

Square & subtract:

✓ **Square root:**

✏ Practice: Word Problems

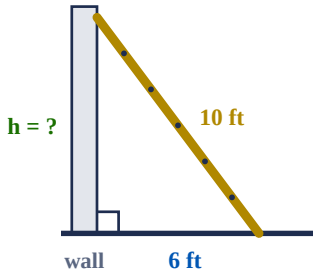
1. ★ Try it A rectangular garden is 15 m long and 8 m wide. A path goes from one corner to the opposite corner. How long is the path?



Answer: _____

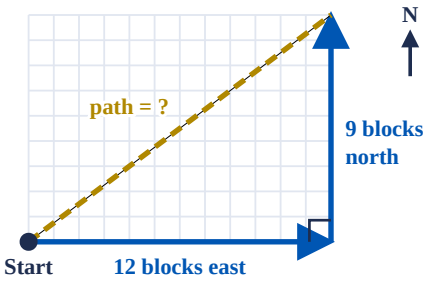
2. ★★ On level A 10-foot ladder leans against a wall. The bottom is 6 feet from the wall. How high up the wall does it reach?

Answer: _____



3. ★★★ Challenge Sam walks 12 blocks east, then 9 blocks north. How many blocks shorter is the straight path back to where he started?

Answer: _____

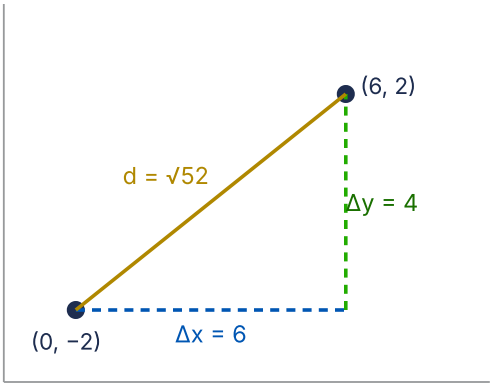


6 Distance on the Coordinate Plane

📐 **HOW IT WORKS:** Connect the two points to form a right triangle. $\Delta x = |x_2 - x_1|$ and $\Delta y = |y_2 - y_1|$. Then use $a^2 + b^2 = c^2$ where $a = \Delta x$, $b = \Delta y$, and c = distance.

⚠️ **NO DISTANCE FORMULA NEEDED!** You already know how to do this — just use the Pythagorean Theorem!

DISTANCE = HYPOTENUSE



Example: Distance from (0, −2) to (6, 2)

Δx:

Δy:

Setup:

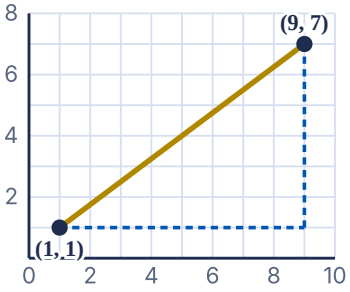
✓ **Answer:**

⚠️ **WATCH YOUR SIGNS:** When subtracting negative coordinates, subtracting a negative means ADDING! Example: 5 − (−3) = 5 + 3 = 8, NOT 5 − 3 = 2.

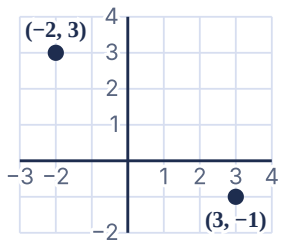
✏ Practice: Distance on the Coordinate Plane

1. ★ Try it Find the distance between (1, 1) and (9, 7).

Answer: _____

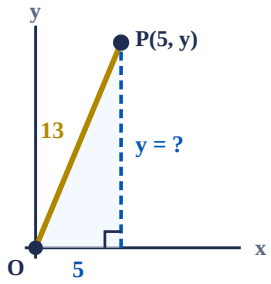


2. ★★ On level Find the distance between (−2, 3) and (3, −1). Round to the nearest tenth (or give the exact √ answer).



Answer: _____

3. ★★★ Challenge Point P is in Quadrant I. It is 13 units from (0, 0), and its x-coordinate is 5. What is its y-coordinate?



Answer: _____

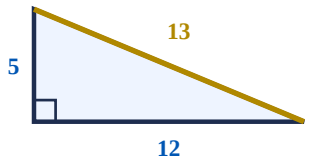
✏ Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

✏ Comprehensive Practice All sections

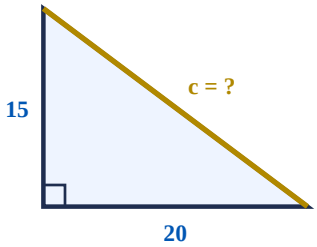
1. ★ Try it A right triangle has legs 5 and 12 and hypotenuse 13. Which equation is true?

- ○ $5^2 + 12^2 = 13^2$
- ○ $5^2 + 13^2 = 12^2$
- ○ $5 + 12 = 13$



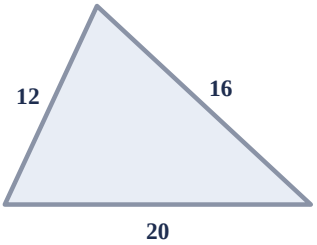
2. ★ Try it The legs of a right triangle are 15 and 20. Find the hypotenuse.

Answer: _____



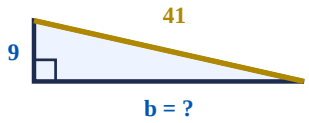
3. ★ Try it A triangle has sides 12, 16, and 20. Is it a right triangle?

- ○ Yes
- ○ No



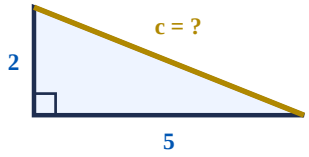
not to scale

4. ★★ On level One leg is 9 and the hypotenuse is 41. Find the other leg.



Answer: _____

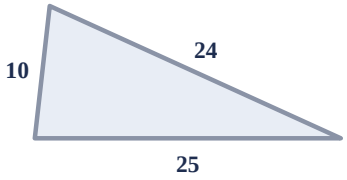
5. ★★ On level The legs are 2 and 5. Find the hypotenuse. Round to the nearest tenth (or give the exact √ answer).



Answer: _____

6. ★★ On level A triangle has sides 10, 24, and 25. Is it a right triangle?

- ○ Yes
- ○ No



not to scale

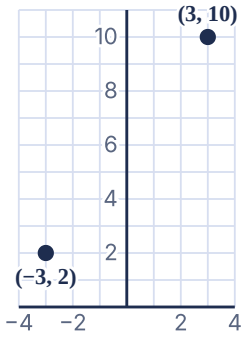
7. ★★ On level A boat sails 7 miles east, then 24 miles north. How far is it from where it started?

Answer: _____



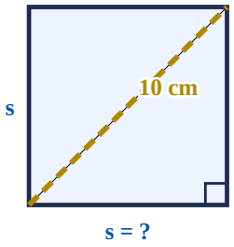
8. ★★ On level Find the distance between (−3, 2) and (3, 10).

Answer: _____



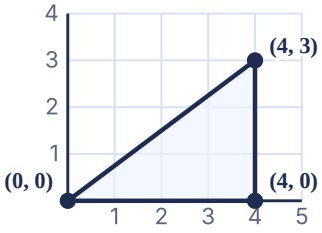
9. ★★★ Challenge A square has a diagonal of 10 cm. How long is each side? Round to the nearest tenth (or give the exact √ answer).

Answer: _____

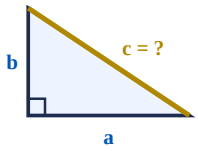


10. ★★★ Challenge A triangle has corners at (0, 0), (4, 0), and (4, 3). What is its perimeter?

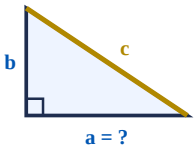
Answer: _____



11. ★★★ Challenge Explain how you know when to ADD the squares and when to SUBTRACT them.



Know: a and b
Find: c



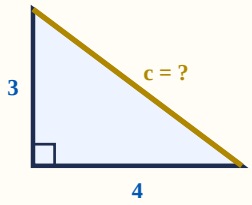
Know: b and c
Find: a

Answer: _____

More Practice by Skill

A. Find the Hypotenuse

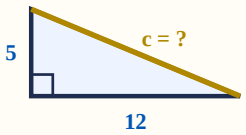
★ Try it **1) Legs 3, 4. c = ?**



c =

_____

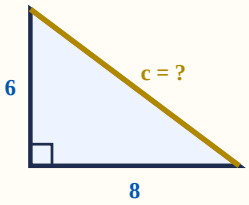
★ Try it **2) Legs 5, 12. c = ?**



c =

_____

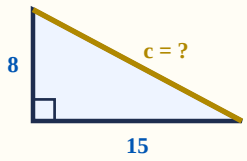
★ Try it **3) Legs 6, 8. c = ?**



c =

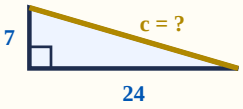
_____

★ Try it 4) Legs 8, 15. c = ?



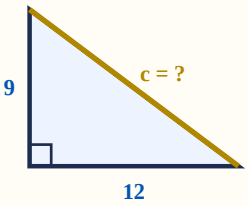
c = _____

★★ On level 5) Legs 7, 24. c = ?



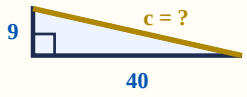
c = _____

★★ On level 6) Legs 9, 12. c = ?



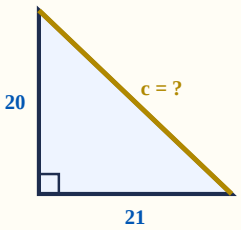
c = _____

★★ On level 7) Legs 9, 40. c = ?



c = _____

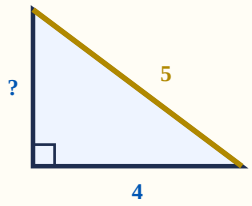
★★ On level 8) Legs 20, 21. c = ?



c = _____

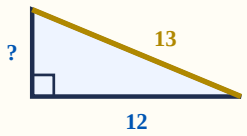
B. Find the Missing Leg

★ Try it 1) Known leg 4, hyp 5. Leg = ?



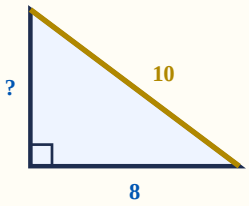
leg = _____

★ Try it 2) Known leg 12, hyp 13. Leg = ?



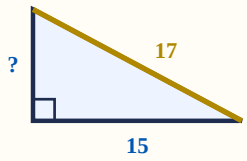
leg = _____

★ Try it 3) Known leg 8, hyp 10. Leg = ?



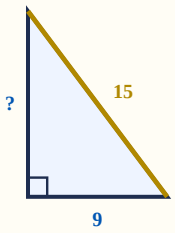
leg = _____

★ Try it 4) Known leg 15, hyp 17. Leg = ?



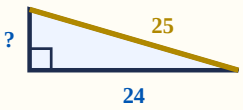
leg = _____

★★ On level 5) Known leg 9, hyp 15. Leg = ?



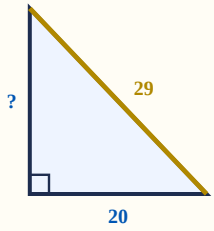
leg = _____

★★ On level 6) Known leg 24, hyp 25. Leg = ?



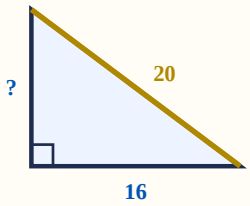
leg = _____

★★ On level 7) Known leg 20, hyp 29. Leg = ?



leg = _____

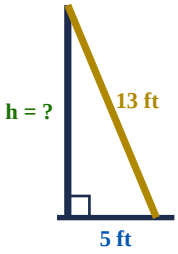
★★ On level 8) Known leg 16, hyp 20. Leg = ?



leg = _____

C. Word Problems

★★★ Challenge **1.** A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high does it reach?

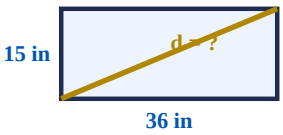


_____

_____

_____

★★ On level **2.** A rectangular TV is 36 inches wide and 15 inches tall. What is the diagonal length?

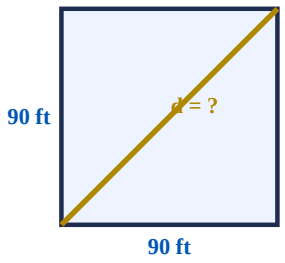


_____

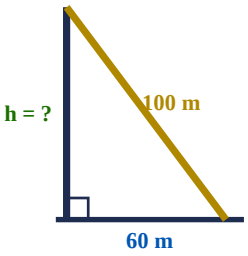
_____

_____

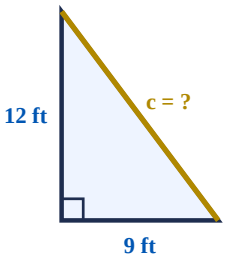
★★★ Challenge **3.** A baseball diamond is a square with 90-ft sides. Distance from home to second base?



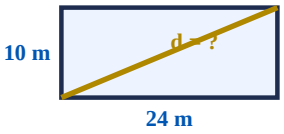
★★★ Challenge **4.** A kite string is 100 m long. The kite is directly above a point 60 m away. How high is the kite?



★★ On level **5.** A support beam covers 9 ft horizontal and 12 ft vertical. How long is the beam?

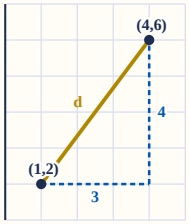


★★ On level **6.** A pool is 24 m long and 10 m wide. A swimmer swims diagonally. How far?



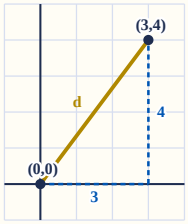
D. Distance on the Coordinate Plane

★ Try it 1) (1,2) → (4,6). d = ?



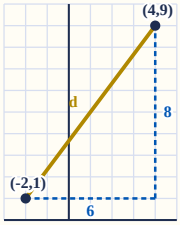
d =

★ Try it 2) (0,0) → (3,4). d = ?



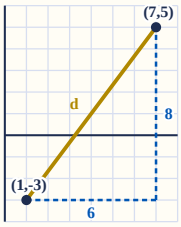
d =

★★ On level 3) (−2,1) → (4,9). d = ?



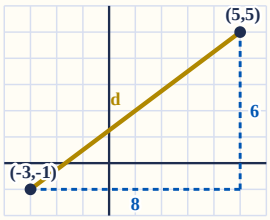
d =

★★ On level 4) (1,−3) → (7,5). d = ?



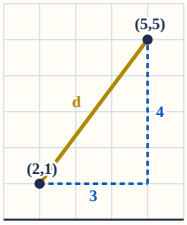
d =

★★ On level 5) (−3,−1) → (5,5). d = ?



d =

★ Try it 6) (2,1) → (5,5). d = ?

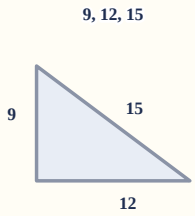


d =

E. The Converse — Is It a Right Triangle?

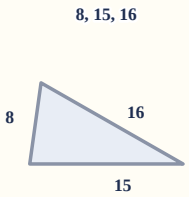
Test the two **smaller** sides against the **largest**. Each figure is drawn to true scale, so do not trust the picture — two of these miss by a hair.

★ Try it 1) Sides 9, 12, 15 — right triangle? (write *yes* or *no*)



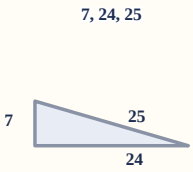
Answer: _____

★★ On level 2) Sides 8, 15, 16 — right triangle? (write *yes* or *no*)



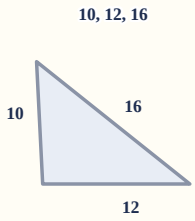
Answer: _____

★ Try it 3) Sides 7, 24, 25 — right triangle? (write *yes* or *no*)



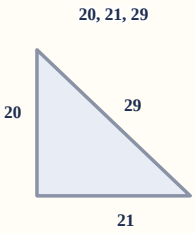
Answer: _____

★★ On level 4) Sides 10, 12, 16 — right triangle? (write *yes* or *no*)



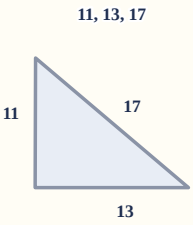
Answer: _____

★★★ Challenge 5) Sides 20, 21, 29 — right triangle? (write *yes* or *no*)



Answer: _____

★★★ Challenge 6) Sides 11, 13, 17 — right triangle? (write *yes* or *no*)



Answer: _____